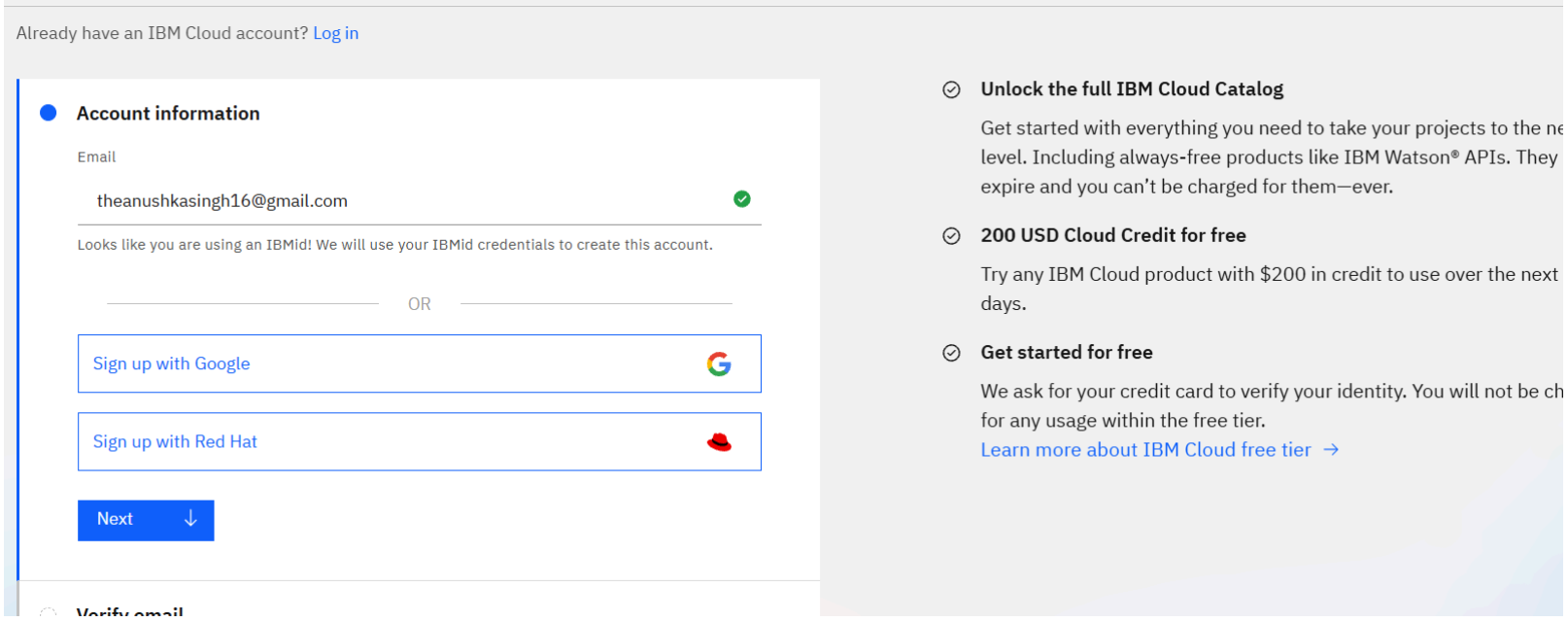
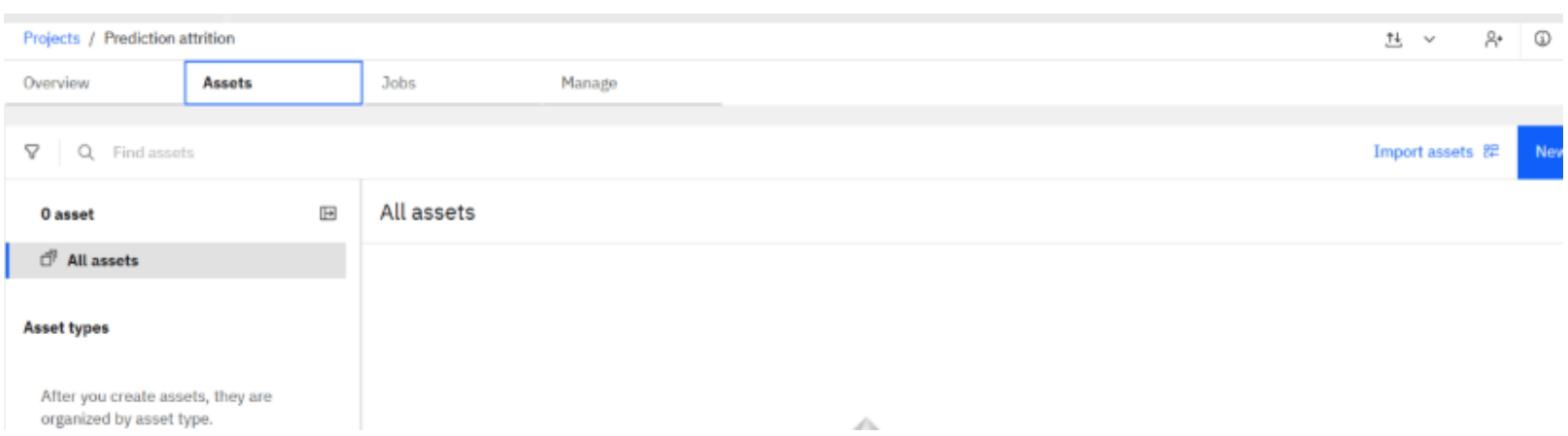
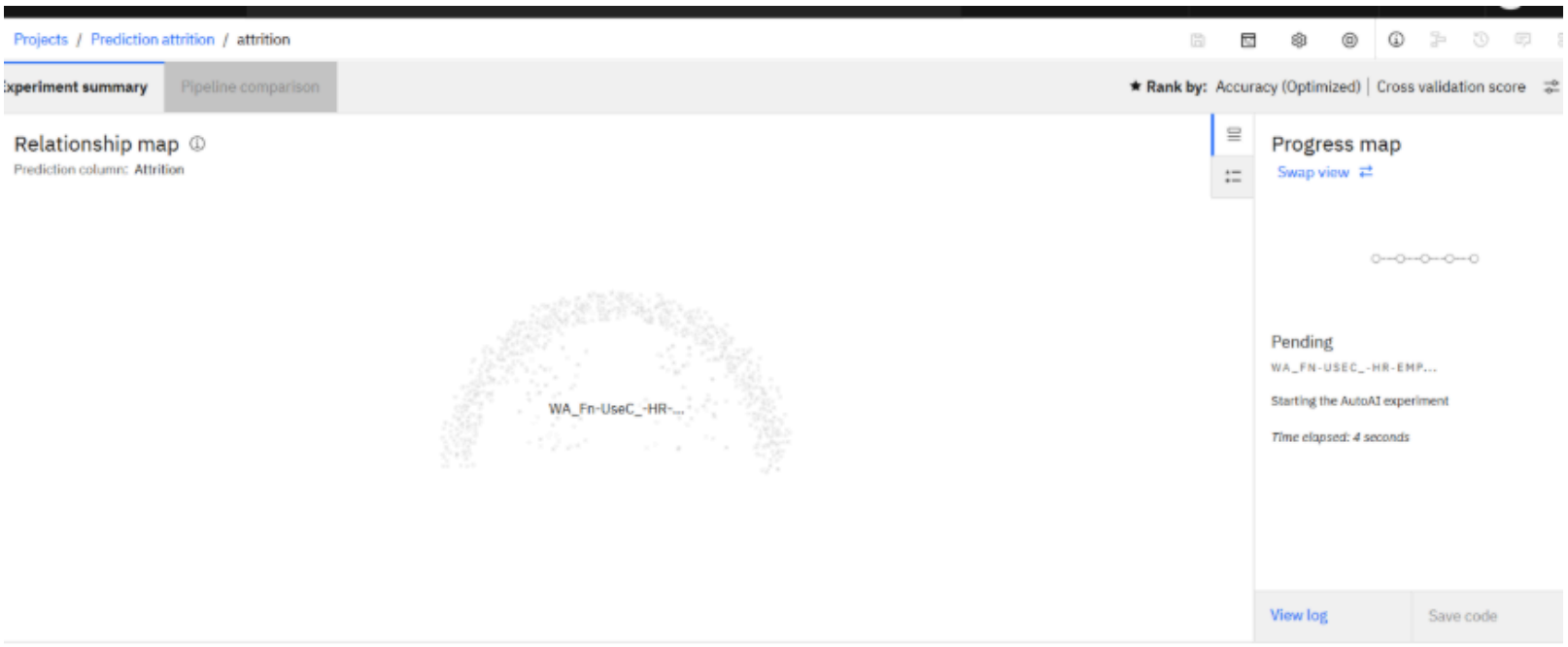
Phase 1: Cleaning Data with Data Refinery

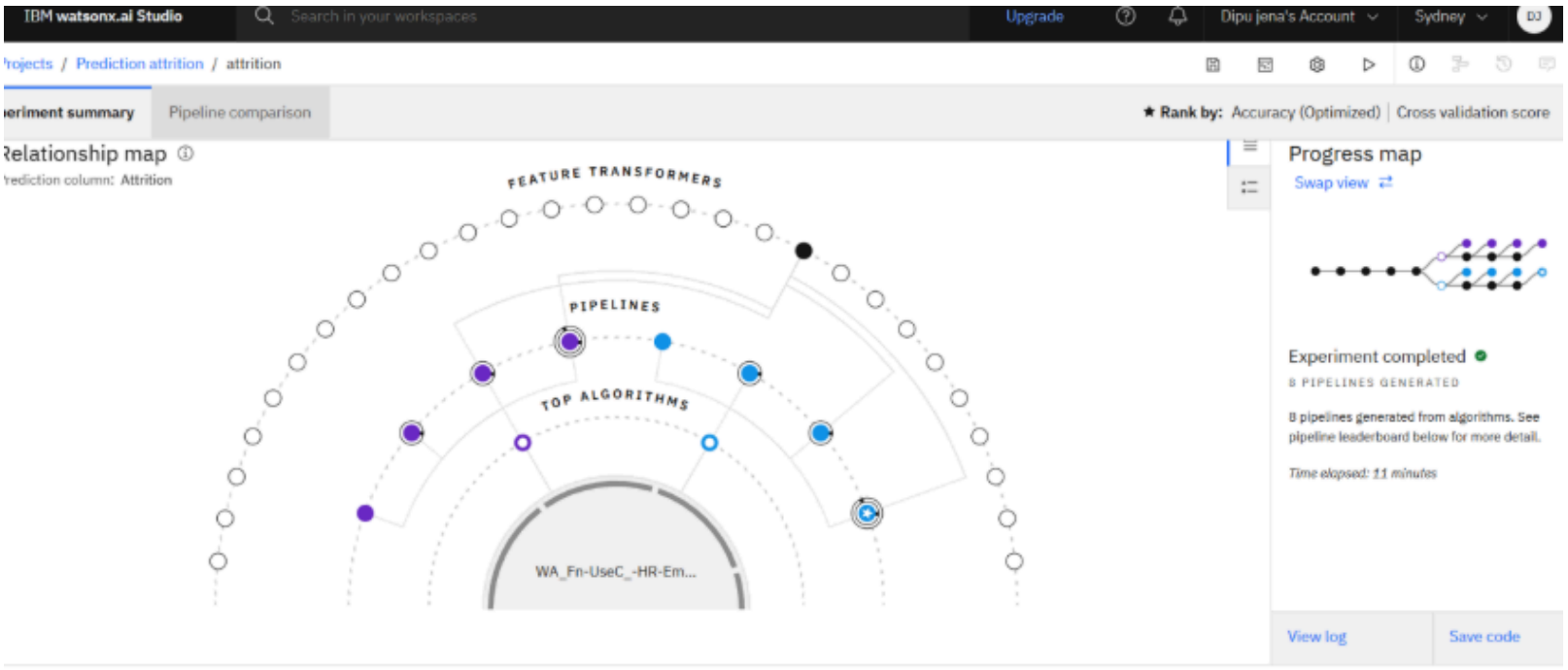
Data Refinery is the "spreadsheet-like" tool for transforming messy data into clean data.

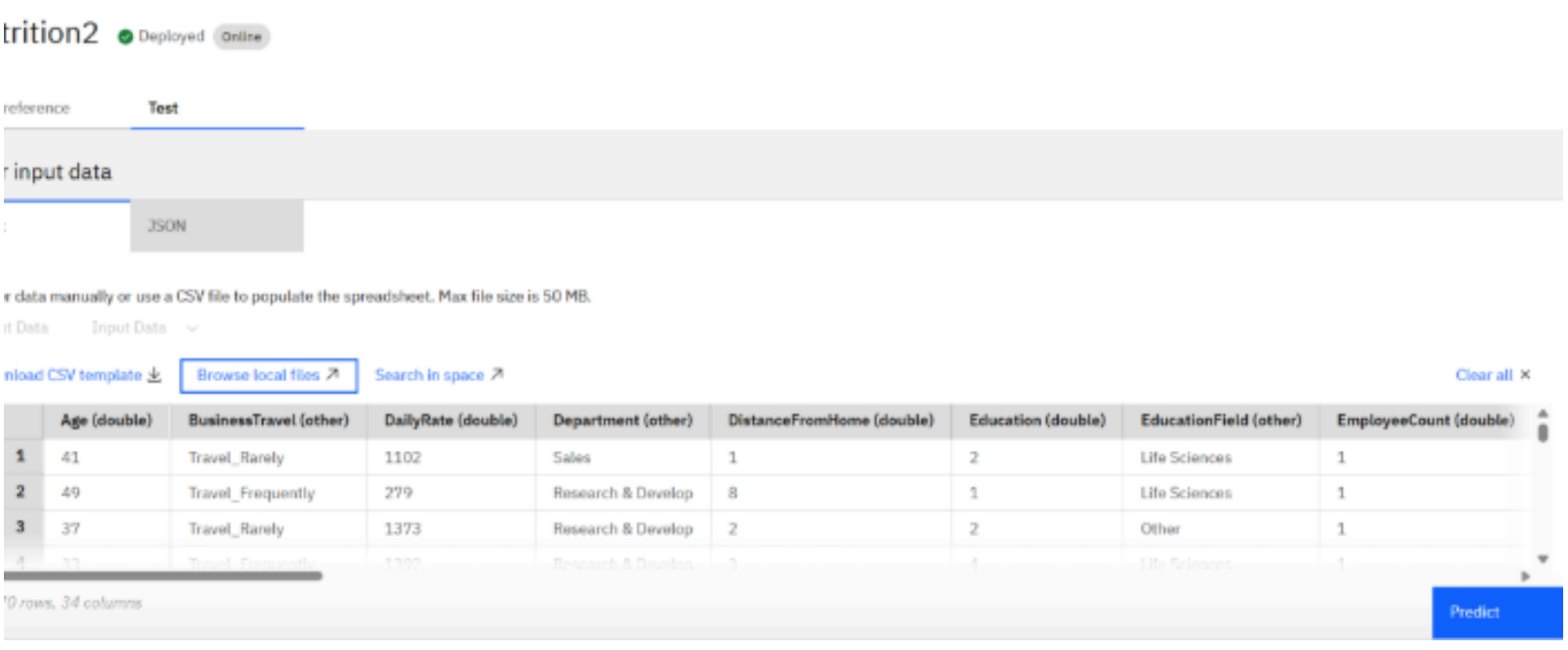
1. **Open Data Refinery:**
    - In your Watson Studio project, go to the **Assets** tab.
    - Click the **three dots (⋮)** next to your uploaded .csv file and select **Refine.**
2. **Apply Transformations:**
    - **Remove Columns:** Click on a column header (like Address or ID) and select **Remove.**
    - **Filter Data:** Click the **+ Operation** button, select **Filter**, and choose a rule (e.g., "Age is greater than 18").
    - **Change Data Type:** Click the icon next to a column name to change a "String" to "Integer" or "Decimal."
3. **Run the Refinery Flow:**
    - Click the **Run** icon (the play button) in the top toolbar.
    - This creates a "Refinery Flow" and saves a new, cleaned version of your dataset (usually named ..._shaped.csv) back to your Assets.



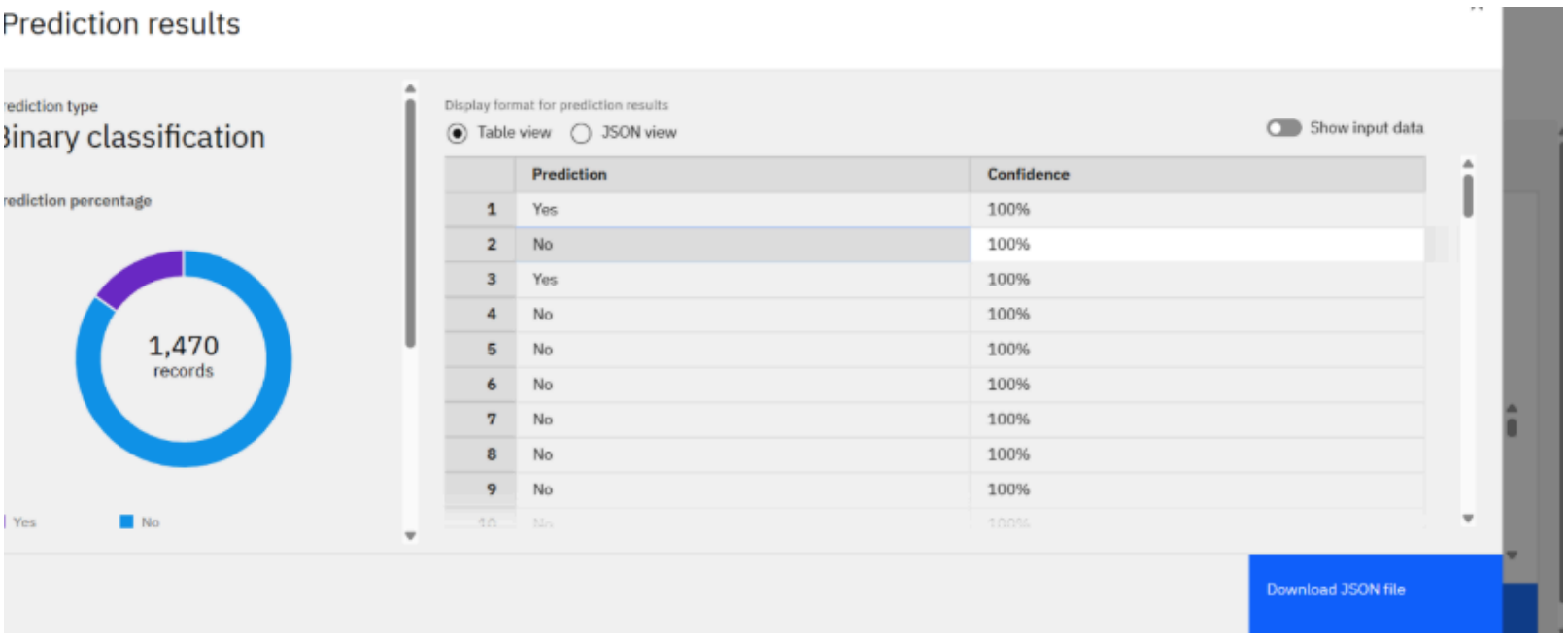








	Age (double)	BusinessTravel (other)	DailyRate (double)	Department (other)	DistanceFromHome (double)	Education (double)	EducationField (other)	EmployeeCount (double)
1	41	Travel_Rarely	1102	Sales	1	2	Life Sciences	1
2	49	Travel_Frequently	279	Research & Develop	8	1	Life Sciences	1
3	37	Travel_Rarely	1373	Research & Develop	2	2	Other	1



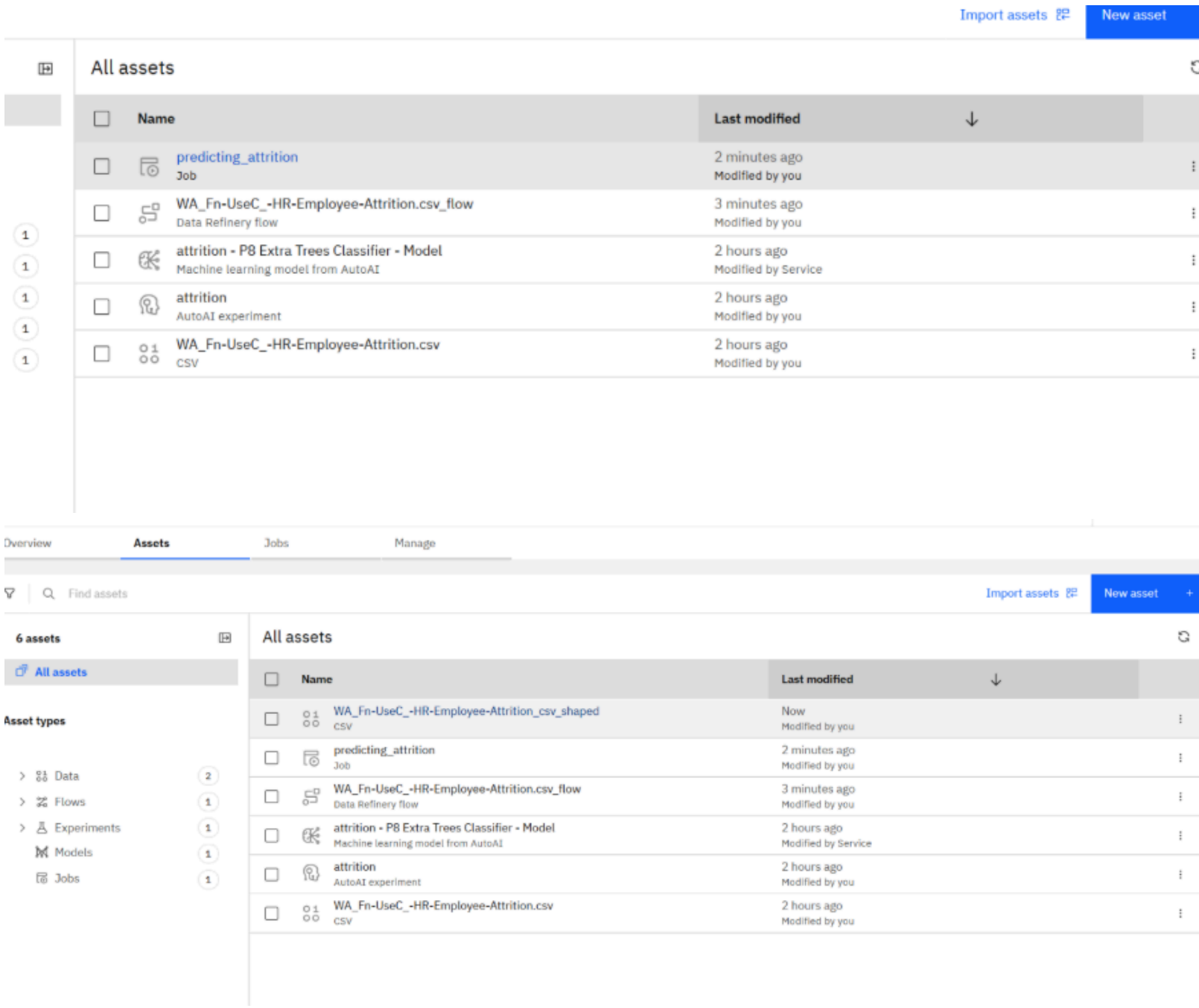
Phase 2: Building Models with AutoAI

AutoAI automates the process of feature engineering, model selection, and hyperparameter tuning.

1. **Launch AutoAI:**
    - Click **New asset** in your project and select **AutoAI**.
    - Give your experiment a name (e.g., *Churn_Prediction_AutoAI*).
2. **Connect Data:**
    - Select the **cleaned/refined** dataset you created in Phase 1.
3. **Configure the Experiment:**
    - **Select Prediction Column:** Choose your target variable (e.g., `Churn` or `Price`).
    - AutoAI will automatically detect if it is a **Classification** or **Regression** problem.
    - Click **Run experiment.**
4. **Analyze the Relationship Map:**
    - Watch the real-time visualization as AutoAI tests different algorithms (Random Forest, XGBoost, etc.) and creates "pipelines."

Phase 3: Evaluating and Saving the Model

1. **Check the Leaderboard:**
    - Once finished, view the **Pipeline leaderboard.** They are ranked by accuracy or error rate.
2. **Inspect a Pipeline:**
    - Click on the top-performing pipeline (e.g., *Pipeline 1*).
    - Review the **Model Evaluation** charts (ROC Curve, Confusion Matrix) and **Feature Importance** (which data columns mattered most).



- Click **Save as** and select **Model.**
- The model is now saved in your project and ready for **Deployment.**